NanoGT Match Report: Mucolipidosis type IV

Disease: Mucolipidosis type IV (ORPHA:578)
Primary gene: MCOLN1
Gene CDS: 1740 bp
Inheritance: Autosomal recessive
Target tissues: CNS, retina

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Libmeldy	LV	9.2/10	[GREEN] High	approved
2	Skysona	LV	8.8/10	[GREEN] High	approved
3	ABO-101	AAV9	8.4/10	[GREEN] High	phase1/2
4	RGX-121	AAV9	8.4/10	[GREEN] High	phase3
5	AVR-RD-01	LV	8.1/10	[GREEN] High	phase1/2

Match #1: Libmeldy

Precedent disease: Metachromatic leukodystrophy
Vector: LV
Tissue target: hematopoietic/CNS
Composite score: 9.2 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.80	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
TOTAL (normalised)	**9.22**	**10.0**	Raw sum / 18 × 10

Rationale

- Gene CDS 1740bp / cargo 8000bp (22% utilized)
- Precedent target match: cns
- Both lysosomal proteins — cross-correction likely
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: leukodystrophy
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Lysosomal enzyme (non-secreted) — moderate cross-correction via M6P pathway; neighbouring cells can benefit but uptake efficiency is lower than fully secreted
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-β2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010

Match #2: Skysona

Precedent disease: Cerebral adrenoleukodystrophy
Vector: LV
Tissue target: hematopoietic/CNS
Composite score: 8.8 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue

Dimension	Score	Max	What it measures
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.80	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
TOTAL (normalised)	**8.78**	**10.0**	Raw sum / 18 × 10

Rationale

- Gene CDS 1740bp / cargo 8000bp (22% utilized)
- Precedent target match: cns
- Both membrane proteins
- LOF inheritance — compatible for gene replacement
- Pathway match: peroxisomal
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Lysosomal enzyme (non-secreted) — moderate cross-correction via M6P pathway; neighbouring cells can benefit but uptake efficiency is lower than fully secreted
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-β2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010

Match #3: ABO-101

Precedent disease: Mucopolysaccharidosis type IIIB
Vector: AAV9

Tissue target: CNS
Composite score: 8.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.80	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
TOTAL (normalised)	**8.39**	**10.0**	Raw sum / 18 × 10

Rationale

- Gene CDS 1740bp / cargo 4700bp (37% utilized)
- Vector tropism plus precedent target match: cns
- Both lysosomal proteins — cross-correction likely
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: lysosomal_storage
- Approval status: phase1/2
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Lysosomal enzyme (non-secreted) — moderate cross-correction via M6P pathway; neighbouring cells can benefit but uptake efficiency is lower than fully secreted
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-β2; minimal T-cell clearance risk

- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010

Match #4: RGX-121

Precedent disease: Mucopolysaccharidosis type II
Vector: AAV9
Tissue target: CNS/liver
Composite score: 8.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.80	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.80	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
TOTAL (normalised)	**8.39**	**10.0**	Raw sum / 18 × 10

Rationale

- Gene CDS 1740bp / cargo 4700bp (37% utilized)
- Vector tropism plus precedent target match: cns
- Both lysosomal proteins — cross-correction likely
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal_storage

- Approval status: phase3
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Lysosomal enzyme (non-secreted) — moderate cross-correction via M6P pathway; neighbouring cells can benefit but uptake efficiency is lower than fully secreted
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-β2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010

Match #5: AVR-RD-01

Precedent disease: Fabry disease
Vector: LV
Tissue target: hematopoietic
Composite score: 8.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.80	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue

Dimension	Score	Max	What it measures
TOTAL (normalised)	**8.11**	**10.0**	Raw sum / 18 × 10

Rationale

- Gene CDS 1740bp / cargo 8000bp (22% utilized)
- No tissue overlap (disease: ['CNS', 'retina'], vector: ['hematopoietic'])
- Both lysosomal proteins — cross-correction likely
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal_storage
- Approval status: phase1/2
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Lysosomal enzyme (non-secreted) — moderate cross-correction via M6P pathway; neighbouring cells can benefit but uptake efficiency is lower than fully secreted
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-β2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010